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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 12/08/2025 |
| **PI Name:** | Fuller, Charles A. Ph.D. | | |
| **Project Title:** | Adaptation of Mouse Systems Physiology to Artificial Gravity via Centripetal Acceleration: Timing, Metabolism, and Aging | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | cafuller@ucdavis.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 530-752-2979 |
| **Organization Name:** | University of California, Davis | | |
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| **City:** | Davis | **State:** | CA |
| **Zip Code:** | 95616-5270 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 08/04/2020 | **End Date:** | 09/03/2025 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Klotz, Rebecca | **Contact Phone:** | 650-604-1119 |
| **Contact Email:** | rebecca.a.klotz@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 09/03/2025 per F. Hernandez/ARC (Ed., 10/8/24).<br>NOTE: End date changed to 09/03/2024 per NSSC information (Ed., 5/17/24).<br>NOTE: End date changed to 05/03/2024 per NSSC information (Ed., 8/18/23). | | |
| **Key Personnel Changes/Previous PI:** | None. | | |
| **COI Name (Institution):** | Hoban-Higgins, Tana Ph.D. ( University of California, Davis )<br>Fuller, Patrick M. Ph.D. ( Unitersity of California, Davis ) | | |
| **Grant/Contract No.:** | 80NSSC20K1501 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | This program was designed to elucidate the dose-response curve for an artificial gravity (AG) exposure countermeasure using an established mouse model and centrifuge aboard the ISS. We proposed to record circadian rhythms of body temperature, food intake, rest/activity, and other video-recorded behaviors during four weeks of AG exposure produced via centrifugation in up to four sequential ISS studies as well as to examine the effects of AG on clock genes underlying circadian function. Additional outcome measures using 'omics analyses (e.g., proteomics and transcriptomics) on select tissues postflight would be used to examine regulatory factors and biomarkers related to our physiological and behavioral measures. Data from subjects exposed to AG will be compared with cohorts studied in microgravity and terrestrial ground controls. This series of studies would be the first to establish a dose-response curve for an AG countermeasure and to answer whether such countermeasures would also be necessary during lunar or Martian habitation. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Exposure to the spaceflight environment adversely affects many aspects of physiology and behavior. Among these deleterious changes are circadian and sleep/wake function alterations, which exacerbate physiological and performance decrements and link to disrupted metabolism and elevated aging biomarkers. Artificial gravity (produced via the centripetal acceleration of centrifugation) has long been proposed as a countermeasure to mitigate or prevent these deleterious changes during spaceflight. It has proved successful in a limited study on mice in the areas of muscle and bone. Planning for a mission to Mars raises the additional question of how effective the Martian gravitational environment will be in combatting microgravity-induced deficits. Finally, for an AG countermeasure to be successfully implemented, the parameters of AG exposure must be experimentally defined. This study will be the first to establish a dose-response curve for an AG countermeasure and to answer whether such countermeasures would also be necessary during lunar or Martian habitation. |
| **Task Progress:** | Compiled from the PI report to the NASA Ames Research Center (Ed., 12/8/25):<br>The MHU-8 spaceflight experiment was successfully developed and implemented. Four groups of mice were housed aboard the ISS within the JAXA Centrifuge facility. [Ed. Note: The Joint Partial-Gravity Rodent Research Mouse Habitat Unit-8 (MHU-8), is a collaboration between NASA, the Japan Aerospace Exploration Agency (JAXA), Harvard Medical School, Northwestern University, Loma Linda University, the University of California-Davis and the University of Tsukuba.] Each group was exposed to a different level of ambient force produced via centrifugation: µG, 0.33G, 0.67G, and 1G. Four ground cage control groups were housed in a flight-like environment, and an additional control group was housed in vivarium conditions. After the flight animals were returned from the International Space Station (ISS), our experimental tissues were collected and appropriately preserved for analysis. During the flight and ground control experiments, video recordings were made to allow for behavioral analysis. These data have also been received. Analysis of the flight and ground control tissues has commenced.<br><br>While data analyses have been ongoing, the liver analyses were initially delayed. We require additional time to complete these analyses and integrate the outcomes across all the variables to interpret the data. The balance of the extension will be utilized to develop and publish the appropriate manuscripts. |
| **Bibliography Type:** | Description: (Last Updated: 12/08/2025) |
| **Articles in Peer-reviewed Journals** | Kremsky I, Pergerson R, Justinen S, Stanbouly S, Willey J, Fuller CA, Takahashi S, Vitaterna MH, Bouxsein M, Mao X. "Artificial gravity attenuates the transcriptomic response to spaceflight in the optic nerve and retina. " Int J Mol Sci. 2024 Nov 9;25(22):12041. https://doi.org/10.3390/ijms252212041 ; PubMed PMID: 39596110; PubMed Central PMCID: PMC11593819 , Nov-2024 |